

Year 9 Pythagoras and Probability Test



Baldivis
Secondary College

Test 4

Name: _____

Class: _____

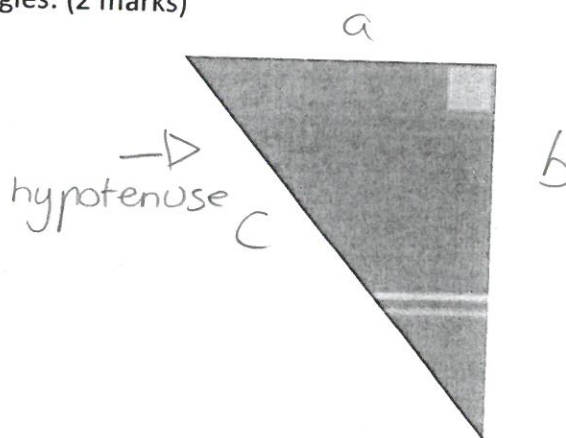
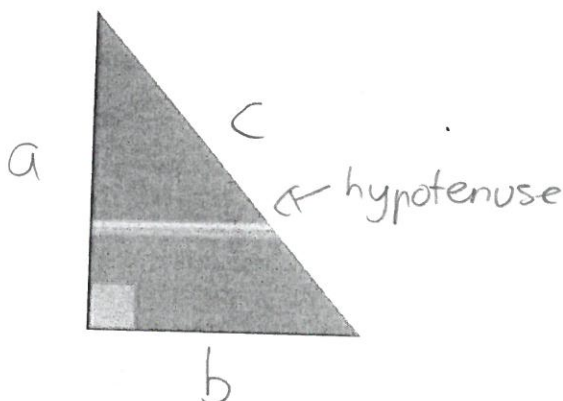
Time: 50 minutes

/45 marks

Resources: Calculator and one page of notes.

Pythagoras Section:

1. Label the hypotenuse on the following triangles: (2 marks)



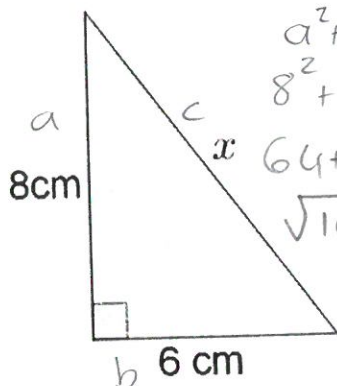
Which is the correct formula for finding the hypotenuse? (1 mark)

- a) $a^2 \times b^2 = c^2$
- ☒ b) $a^2 + b^2 = c^2$
- d) $c^2 - b^2 = a^2$

2. Which is the correct formula for finding a short side? (1 mark)

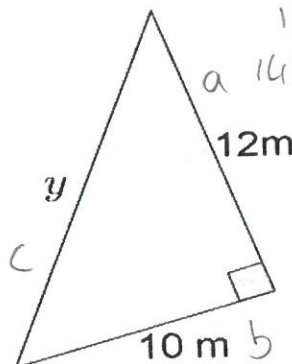
- a) $a^2 + b^2 = c^2$
- b) $c^2 \times b^2 = a^2$
- ☒ d) $c^2 - b^2 = a^2$

Find the length of the pronumeral in the following triangles. Give your answer correct to two decimal places if required. (9 marks)



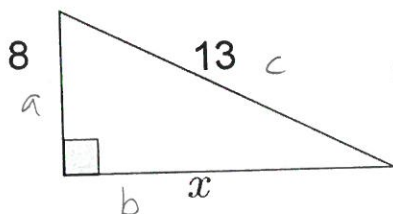
$$\begin{aligned} a^2 + b^2 &= c^2 \\ 8^2 + 6^2 &= c^2 \\ 64 + 36 &= 100 \\ \sqrt{100} &= \underline{10\text{cm}} \end{aligned}$$

work
Root
Ans



$$\begin{aligned} a^2 + b^2 &= c^2 \\ 12^2 + 10^2 &= c^2 \\ 144 + 100 &= 244 \\ \sqrt{244} &= 15.62\text{m} \end{aligned}$$

work
Root
Ans



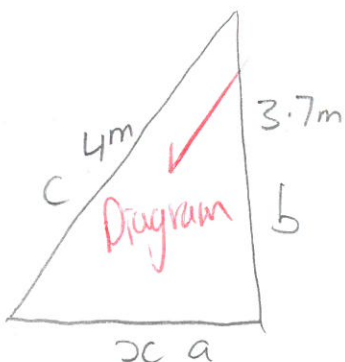
$$\begin{aligned} c^2 - b^2 &= a^2 \\ 13^2 - 8^2 &= b^2 \\ 169 - 64 &= 105 \\ \sqrt{105} &= 10.25 \end{aligned}$$

formula
subtract
Ans

9

3. A 4 m tall ladder reaches 3.7 m up a wall. Find the horizontal distance between the base of the wall and the ladder. Give your answer correct to two decimal places. (hint: Draw a picture)

[4 marks]



$$\begin{aligned} c^2 - a^2 &= b^2 \\ 4^2 - 3.7^2 &= b^2 \\ 16 - 13.69 &= 2.31 \\ \sqrt{2.31} &= 1.52\text{m away from the base of the wall.} \end{aligned}$$

Subtracts

Substituted

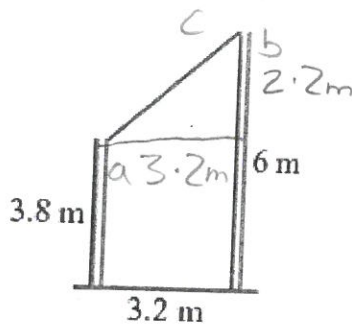
Answer

4

13

4. Two poles are located 3.2 m apart. A wire links the tops of the two poles. Find the length of the wire if the poles are 3.8 m and 6 m in height. Give your answer correct to one decimal place.

[3 marks]



$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 3.2^2 + 2.2^2 &= c^2 \\
 10.24 + 4.84 &= 15.08 \\
 \sqrt{15.08} &= 3.88\text{m} \rightarrow \text{length of wire}
 \end{aligned}$$

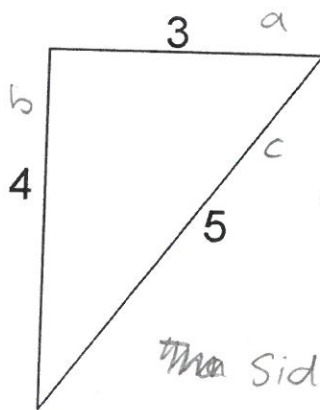
Ans

5. Are the following right-angled triangles. Justify your answer.

a.

b.

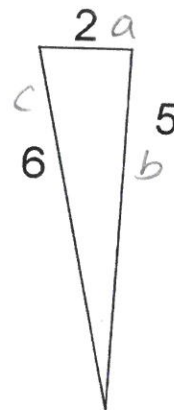
[4 marks]



$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 3^2 + 4^2 &= c^2 \\
 9 + 16 &= 25 \\
 \sqrt{25} &= 5
 \end{aligned}$$

Side A² + Side B² = Side C²

any reasonable answer.
2 marks



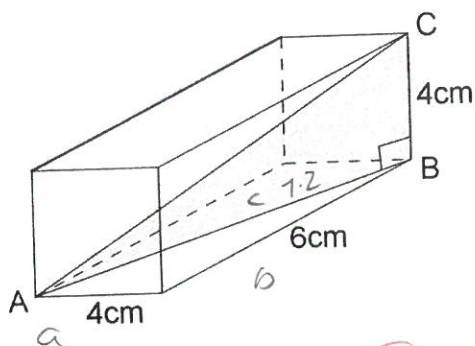
$$\begin{aligned}
 a^2 + b^2 &= c^2 \\
 2^2 + 6^2 &= c^2 \\
 4 + 36 &= 40 \\
 \sqrt{40} &= 6.3
 \end{aligned}$$

$a^2 + b^2$ doesn't equal c^2 Not a Right angle

I just know it

6. Consider the following rectangular prism.

[4 marks]



a. Find the length AB as an exact value.

$$\begin{aligned} a^2 + b^2 &= c^2 \\ 4^2 + 6^2 &= c^2 \\ 16 + 36 &= 52 \\ \sqrt{52} & \end{aligned}$$

b. Hence, find the length AC. Round your answer to two decimal places.

$$\begin{aligned} 4^2 + 7.2^2 &= c^2 \\ 16 + 51.84 &= c^2 \\ \sqrt{67.84} & \end{aligned} \quad 8.24m$$

Probability Section:

7. A coin is flipped ^{once} what is the probability of getting: (2 marks)

a) A head = $\frac{1}{2}$

b) 2 tails = ~~A~~ No chance.

8. Ken has 20 marbles in a bag. 12 are green and 8 are red. He selects a marble at random. What is the probability of the following: (4 marks)

Ken selects a green marble: $P = \frac{12}{20} = \frac{3}{5}$

Ken selects a red marble: $P = \frac{8}{20} = \frac{2}{5}$

He selects a white marble: $P = \frac{0}{20}$

He selects a marble: $P = \frac{20}{20}$

- Mary
- M
 - W
 - CC
 - G
 - B
 - S
 - D
 - CC
 - B
 - V
 - CC
 - B
 - S
- Handwritten notes on the right side of the diagram:
- cut ones
 - GC
 - GB
 - GS
 - DC
 - DS
 - DB
 - VC
 - VS
 - VB
 - VV
 - any
 - missy

$$\frac{1}{9}$$
$$\frac{7}{9}$$

-
- Win Lawnbowls
- lose
- Win table tennis
- Lose table tennis
- Win
- lose
- lawnbowls
- lose

a) What is the probability of Jackson losing both lawn bowls and table tennis?

$$\frac{5}{9} \times \frac{2}{6} = \frac{10}{54}$$

$$0.55 \times 0.33 = 0.1815$$

$$0.1815 \times 100 = 18.15\%$$

End of test